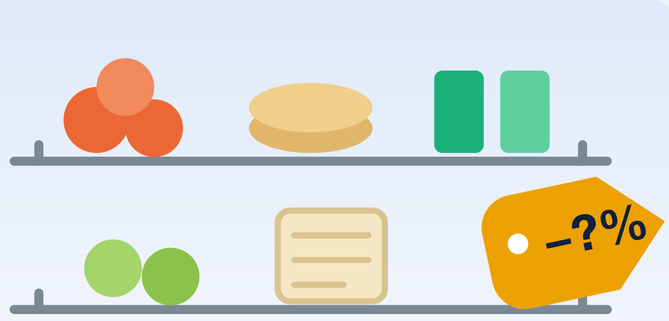
HKU MAIB 7002 · GROUP Q · MACHINE LEARNING FOR PRICING

Cut the price, or trust the shelf?

We taught a model to predict what a perishable will sell tomorrow at any discount. Then we discovered it would discount everything, and had to teach it to read the shelf first.

Aashish Omprakash Pareek · Yuanyuan Yin · Lingyu Chen · Dingdong FreshRetailNet-50K, a public dataset of 50,000 store-product histories



6 pm · the shelf is still half full · what is tomorrow's price?

01 A decision every evening, for every product

6 pm, the shelf is half full

Tomorrow's discount must be set tonight, before anyone knows how much will sell.

Two ways to lose

Hold the price and stock may be thrown away. Cut it and margin goes on sales that would have happened anyway.

Our question

For one product, in one store, on one day: how much sells tomorrow at each discount, and should we discount at all?

02 The raw material

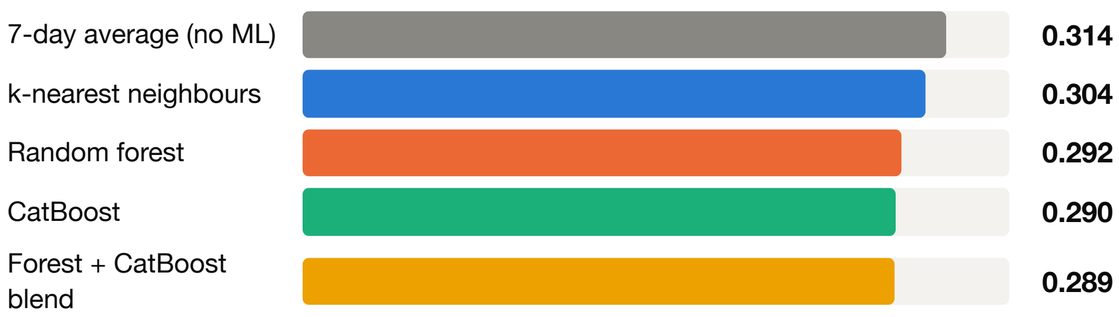
312
product histories in five stores, picked because their discounts vary

28,080
store-product-days; one unseen week of 2,184 kept for the test

45%
of days have an empty-shelf hour, a clue we later put to work

03 Four models, one unseen week

Each model learns from the product's own recent sales, today's stock and tomorrow's planned discount. Error on the 2,184 held-out days; lower is better.



What we learned: every model beat the no-ML average, the three tree models tie within noise, and choosing the inputs mattered more than any tuning dial. Dropping the weather was the single biggest gain.

04 The decisions behind the numbers

k-nearest neighbours · 25 look-alike days
Finds the 25 most similar past days and averages them. Simple, and already 5% better than the average.

Random forest · 300 trees, depth 10
All 16 settings we tried beat the baseline every single week. A forgiving model.

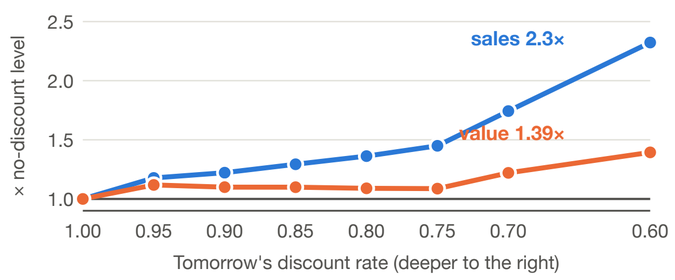
CatBoost · small and slow on purpose
Boosting over-fits in plain sight (see 06), so the best version is 500 shallow trees at a 0.03 learning rate.

Blend · 50/50 forest + CatBoost
Their mistakes are 97% correlated, so blending gains only 0.001. We keep it as a safe choice, not a clever one.

Splits are strictly in time order: every validation week is predicted by a model that has never seen it. Random splits would have flattered every model by putting near-identical days on both sides.

05 The twist: a good forecaster is a reckless pricer

Ask the model "what if I discount?" and it answers: deeper is always better. At a 0.6 rate it predicts 2.3× the sales.



It learned that from clearance days, which are fewer than 1% of the data. Left alone it would discount on 98% of days.

So we put the shelf before the model. Three stock rules decide first; the model only chooses among discounts the product has actually run.

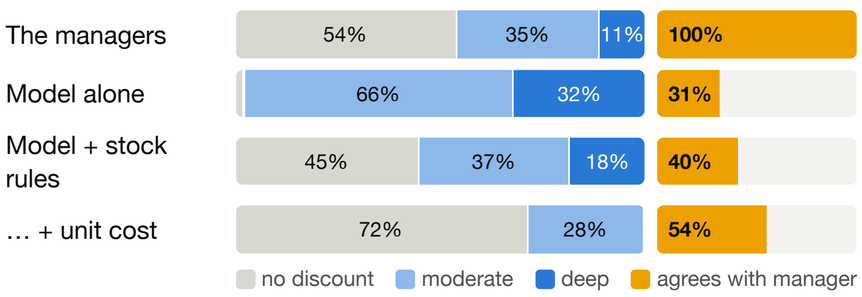
Sold out today?
Then no discount tomorrow: the shelf is clear.

A week with no sell-out?
Then a discount is required; the model picks how deep.

Otherwise ask the model
Only about discounts this product has actually run.

Choose
Highest predicted value, net of unit cost, if the gain is worth it.

What each policy recommends on the test week, and how often it lands in the same band as the real managers.



Stock rules decide 53% of days before the model is asked. Agreement rises from 31% to 40%, and to 54% once a unit cost makes deep discounts cost something.

06 The challenges we found most interesting

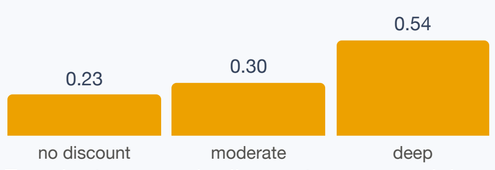
The shelf lies
On 45% of days the shelf was empty for part of the day, so recorded sales are lower than real demand. The model learns from censored numbers.

Sales without units or prices
The data is normalised: 1.2 is not 1.2 kilograms or dollars. "Value" is sales × discount rate, a proxy, so profit cannot be computed.

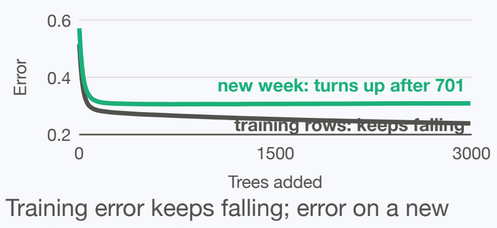
Managers are not random
Discounts were chosen on days managers expected demand. Part of the lift the model sees is their judgement, not the price cut. Correlation, not cause.

No counterfactual
We only ever see what sold at the price the manager chose. Whether our policy sells more or wastes less needs a real experiment.

Weather made things worse
0.327 → 0.312
Removing rain, temperature and humidity cut the error more than any amount of tuning did.

Deep discounts are the hard days


Error by tomorrow's discount: every model under-predicts the spike on exactly the days a pricing tool cares about.

Boosting over-fits in plain sight


Training error keeps falling; error on a new week turns up after 701 trees.

One city, one week, one guess at shelf life
Five stores in one city and a single test week that has been scored for four models. The "no sell-out for a week means old stock" rule assumes a shelf life the data never states.

Data: Dingdong-Inc/FreshRetailNet-50K on Hugging Face, CC BY 4.0 · scikit-learn and CatBoost · Errors are mean absolute error in normalised sales units

Full notebook, code and Docker image: github.com/Lisayinyy/MAIB_7002, branch pricing-ml-ap